SOFTWARE ENGINEERING

LAB 1 | Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #20

Hospital Bed & ICU Allocation Dashboard

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Problem Statement • Requirements Table • UML Use-Case Diagram • Use-Case Flow Specification

Problem Statement

1. Problem Context & Overview

An emergency room bed management dashboard displaying real-time ward occupancy, prioritizing ICU transfers based on triage score, and orchestrating inter-facility ambulance transfer workflows.

2. Target Stakeholders / Actors

Triage Nurse	Hospital Administrator

3. Sample Functional Requirement

FR-001 [Priority: High]
Description: The system shall compute triage severity scores from intake vitals and dynamically order the ER bed allocation waitlist.
Sample Acceptance Criteria: Pass: Higher acuity patient automatically moves ahead of lower acuity patient; Fail: FIFO queue ignores critical emergency.

4. Sample Non-Functional Requirement

NFR-001 [Type: Performance & Security]
Description: Bed status transitions (Occupied, Cleaning, Available) must propagate to all connected ER dashboard clients in under 1 second via WebSockets.
Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

5. Required Deliverables

Complete Requirements Table — exactly 5 Functional Requirements (FR-001 to FR-005).

2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

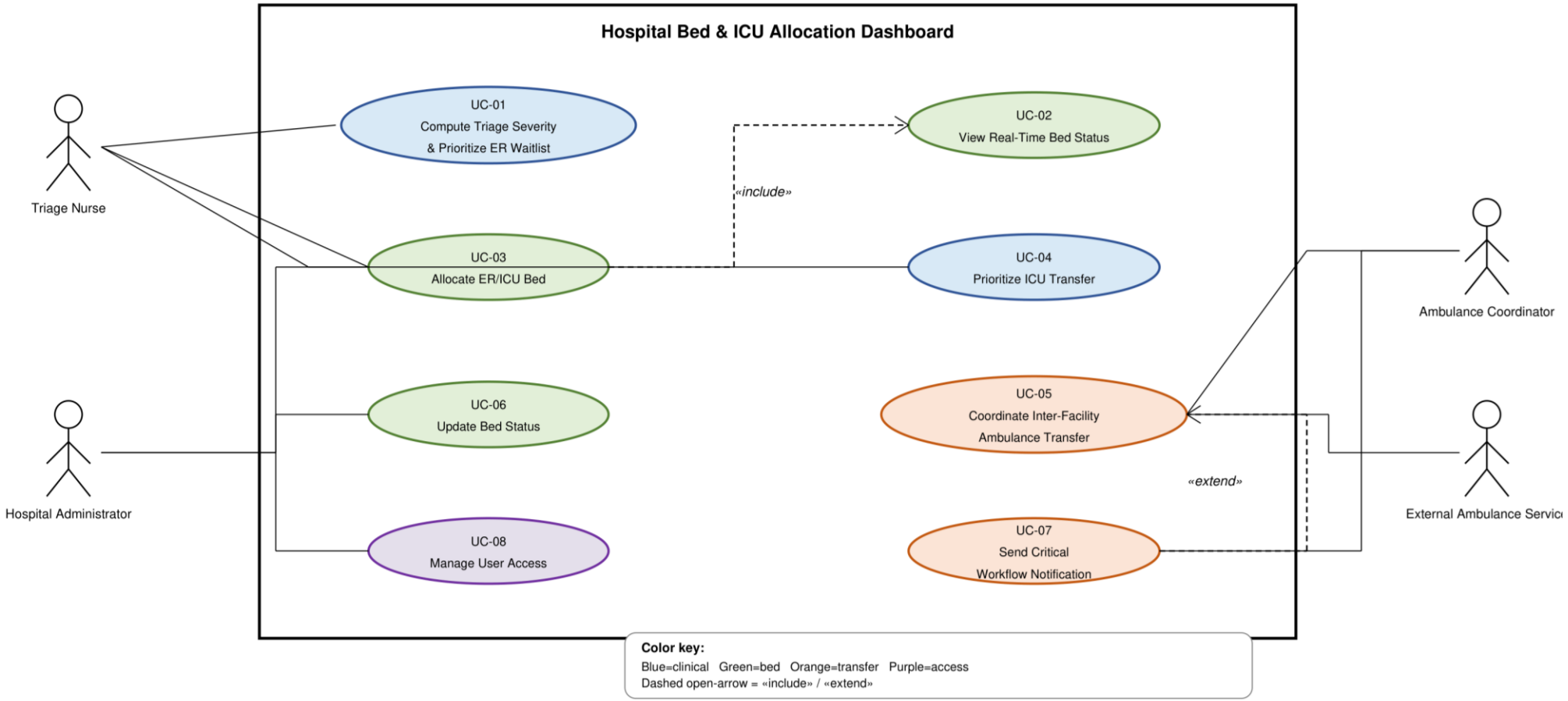
UML Use-Case Diagram — model all actors and primary use cases and include at least one «include» and one «extend» relationship.

Use-Case Flow Specification — 1-page document for one core use case with Preconditions, Postconditions, Main Success Scenario, and one Alternate Flow.

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall compute triage severity scores from intake vitals and dynamically order the ER bed allocation waitlist.	High	Pass: When a patient's intake vitals are submitted, the system calculates a severity score and places the patient in the waitlist according to acuity. Fail: The waitlist remains FIFO when a higher-acuity patient is entered.	Core requirement for prioritizing emergency patients and allocating scarce ER beds according to triage severity.	Given in problem statement; wording retained with measurable acceptance criterion.
FR-002	Functional	The system shall display the real-time status of beds by ward, including Occupied, Cleaning, and Available, and allow authorized staff to update a bed status.	High	Pass: After an authorized user changes a bed status, the new status is displayed for that bed and ward. Fail: The old status remains displayed or an unauthorized user can change it.	Provides the operational visibility required for real-time bed and ICU allocation.	Status values are restricted to those specified by the problem statement.
FR-003	Functional	The system shall support ICU transfer prioritization by presenting eligible patients in triage-score order together with currently available ICU beds.	High	Pass: For two eligible patients, the patient with the higher triage severity appears first, and the dashboard shows the current available ICU beds. Fail: The lower-severity patient is ranked first or bed availability is missing.	Directly supports the stated requirement to prioritize ICU transfers based on triage score and bed availability.	Designed to make the allocation decision traceable and consistent.
FR-004	Functional	The system shall initiate and track an inter-facility ambulance transfer workflow, including transfer request, ambulance assignment, dispatch, and completion status.	High	Pass: A transfer can progress through Request → Assigned → Dispatched → Completed and each status is visible to authorized users. Fail: A required status cannot be recorded or the transfer cannot be tracked.	Supports the stated orchestration of inter-facility ambulance transfer workflows.	Status sequence is defined for testability.
FR-005	Functional	The system shall notify relevant authorized staff when a critical bed allocation, ICU transfer, or ambulance transfer status requires their attention.	Medium	Pass: When a configured critical event occurs, the responsible authorized user receives a dashboard notification for that event. Fail: No notification is generated for the configured event.	Reduces delays in emergency coordination by making important workflow changes visible to responsible staff.	Notification is limited to relevant workflow events rather than every dashboard change.
NFR-001	Nonfunctional	Bed status transitions (Occupied, Cleaning, Available) must propagate to all connected ER dashboard clients in under 1 second via WebSockets.	High	Pass: Benchmarking tests confirm that a bed-status update reaches all connected dashboard clients in < 1 second under the defined simulated peak load. Fail: The measured propagation time is ≥ 1 second or the update is not delivered to a connected client.	Given in problem statement and essential for a real-time emergency dashboard.	Performance target retained from the problem statement.
NFR-002	Nonfunctional	The system shall enforce role-based access control so that only authenticated users with the required role can perform protected triage, bed-allocation, administration, or ambulance-transfer actions.	High	Pass: In security testing, 100% of unauthorized protected-action attempts are denied, while authorized test users can access only the functions assigned to their role. Fail: Any unauthorized protected action succeeds.	Protects patient-related operational data and prevents unauthorized changes to emergency allocation and transfer workflows.	Roles modeled in this assignment include Triage Nurse and Hospital Administrator; operational ambulance coordination is also represented.

UML Use-Case Diagram



Use-Case Flow Specification

UC-04 — Prioritize ICU Transfer

Primary Actors	Triage Nurse; Hospital Administrator
Supporting Actors	Ambulance Coordinator; External Ambulance Service
Preconditions	Patient has a recorded triage severity score; ICU bed-status data is available; user is authenticated and authorized.
Postconditions	The prioritized ICU-transfer decision is recorded/displayed; if transfer is initiated, the ambulance workflow can be tracked.

Main Success Scenario

- Triage Nurse selects the ICU transfer prioritization function.
- System retrieves eligible patients and their current triage severity scores.
- System retrieves the current ICU bed availability/status.
- System orders eligible patients by triage severity, placing higher-acuity patients ahead of lower-acuity patients.
- System displays the prioritized transfer list together with available ICU beds.
- Triage Nurse selects the highest-priority eligible patient for transfer.
- System records the ICU transfer prioritization decision and updates the workflow status.
- System displays the updated transfer state to authorized users.

Alternate Flow — ICU Bed Unavailable

- System finds that no ICU bed is currently available.
- System displays the patient's priority position and the current ICU availability as unavailable.
- System keeps the patient in the prioritized transfer list and allows authorized staff to monitor the bed-status change.
- When an ICU bed becomes available, the system updates the dashboard and the patient can be considered for transfer again according to the current priority order.

Traceability note: This flow operationalizes FR-001, FR-002, and FR-003 and uses the real-time bed-status behavior specified by NFR-001.